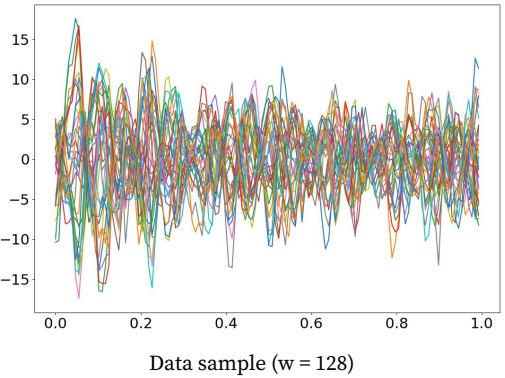
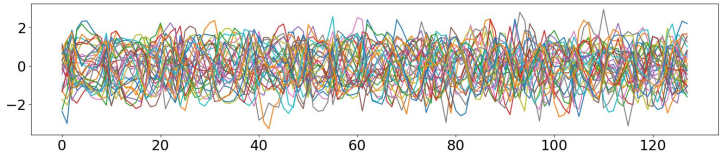


Data sample (w = 128)

Choose one

a) Raw signal

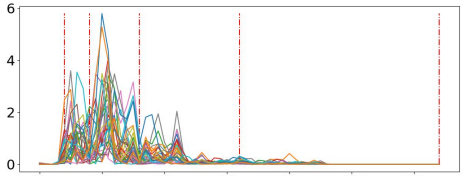
Z-score normalization →



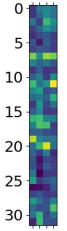
Features w/ shape (32, w)

b) Power Spectral Density

Estimate Power Spectral Density using Welch's method and split into frequency bands →

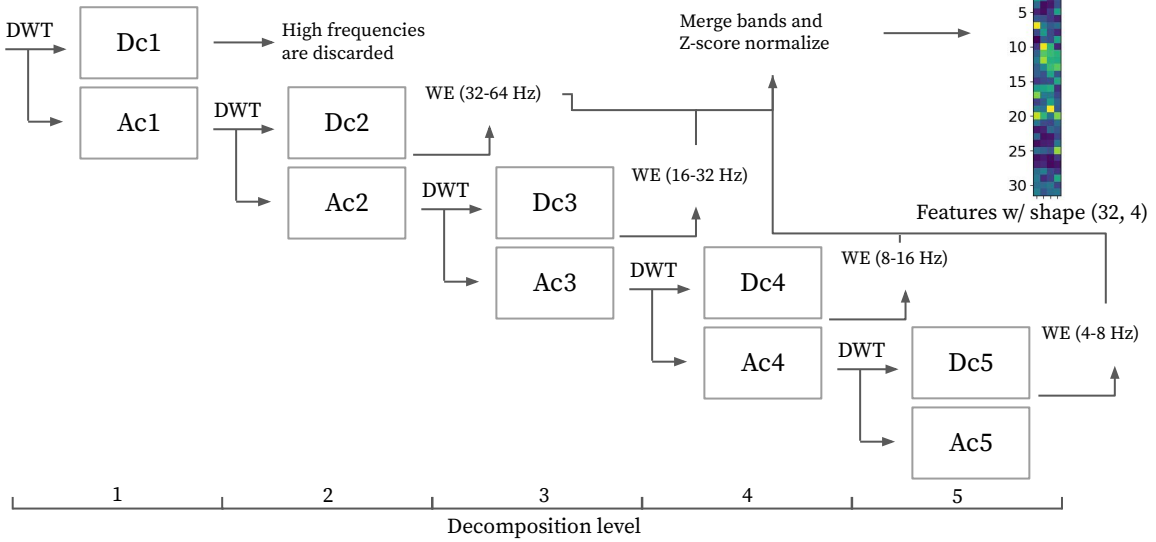


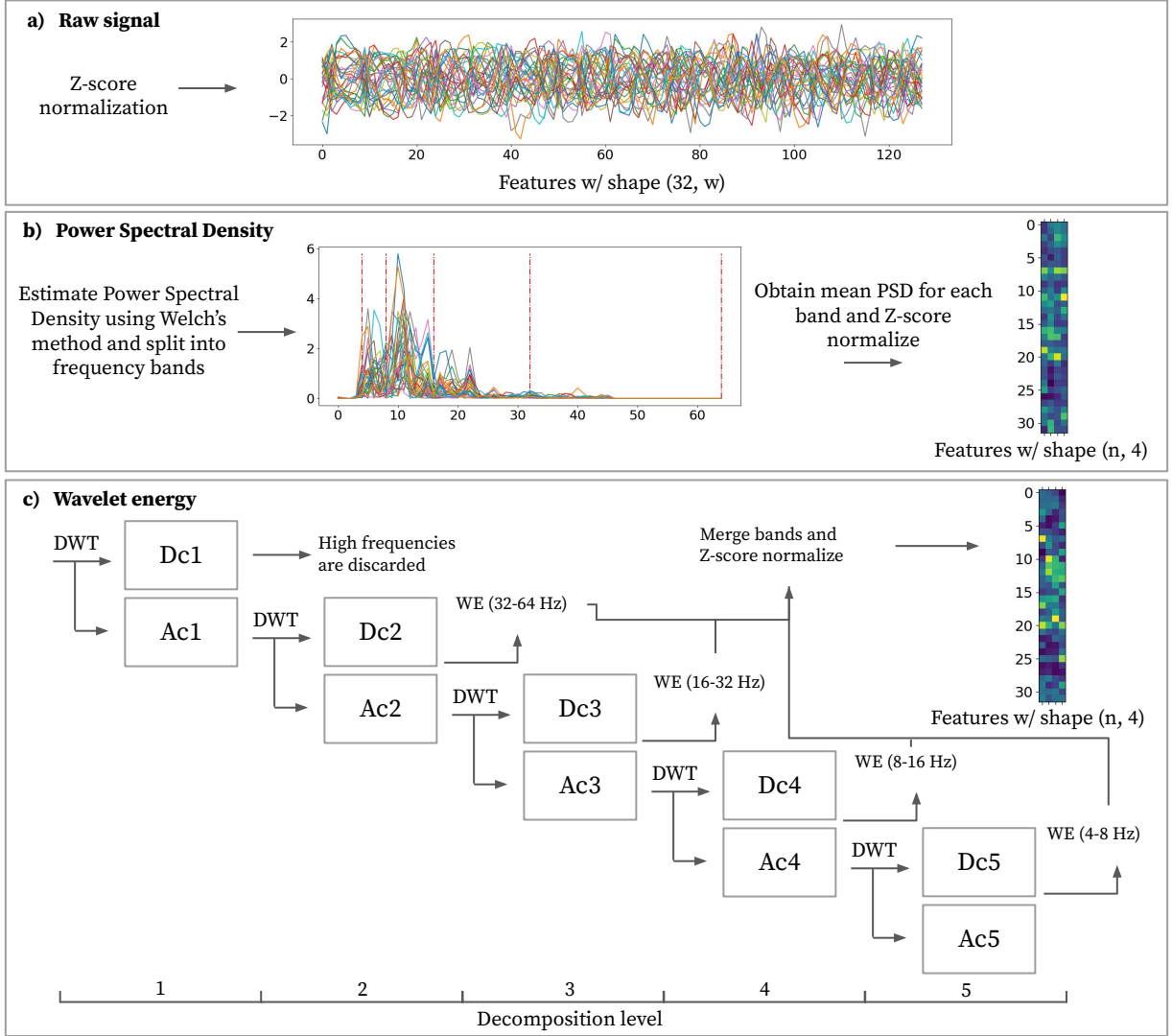
Obtain mean PSD for each band and Z-score normalize →

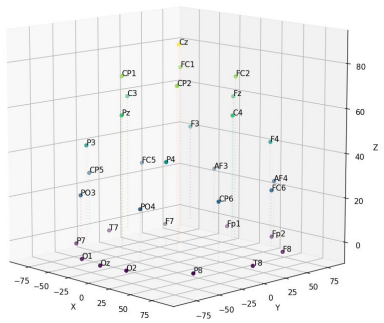


Features w/ shape (32, 4)

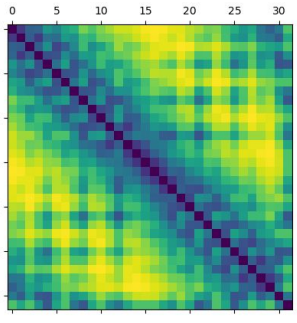
c) Wavelet energy



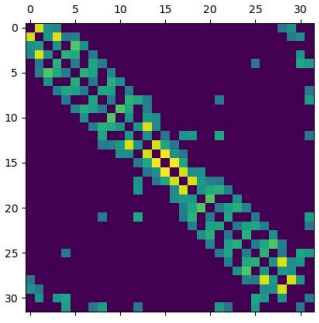

a) Raw signal
Z-score normalization
Features w/ shape (32, w)
b) Power Spectral Density
Estimate Power Spectral Density using Welch's method and split into frequency bands
Obtain mean PSD for each band and Z-score normalize
Features w/ shape (n, 4)
c) Wavelet energy
DWT
Dc1
Ac1
High frequencies are discarded
Dc2
Ac2
Dc3
Ac3
Dc4
Ac4
Dc5
Ac5
WE (32-64 Hz)
WE (16-32 Hz)
WE (8-16 Hz)
WE (4-8 Hz)
Merge bands and Z-score normalize
Features w/ shape (n, 4)
Decomposition level



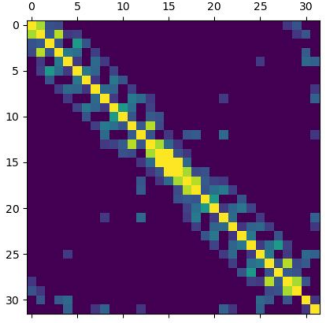
a) Electrode positions



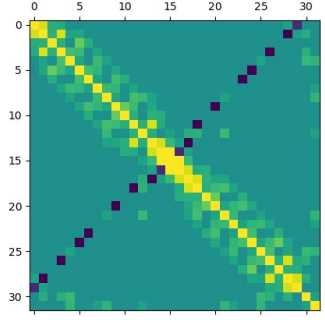
b) Distance matrix



c) Local connections



d) Added self-loops



e) Global connections

(1) $$\mathbf{A}_{ij} = \min(1, \frac{\delta}{d_{ij}^2})$$

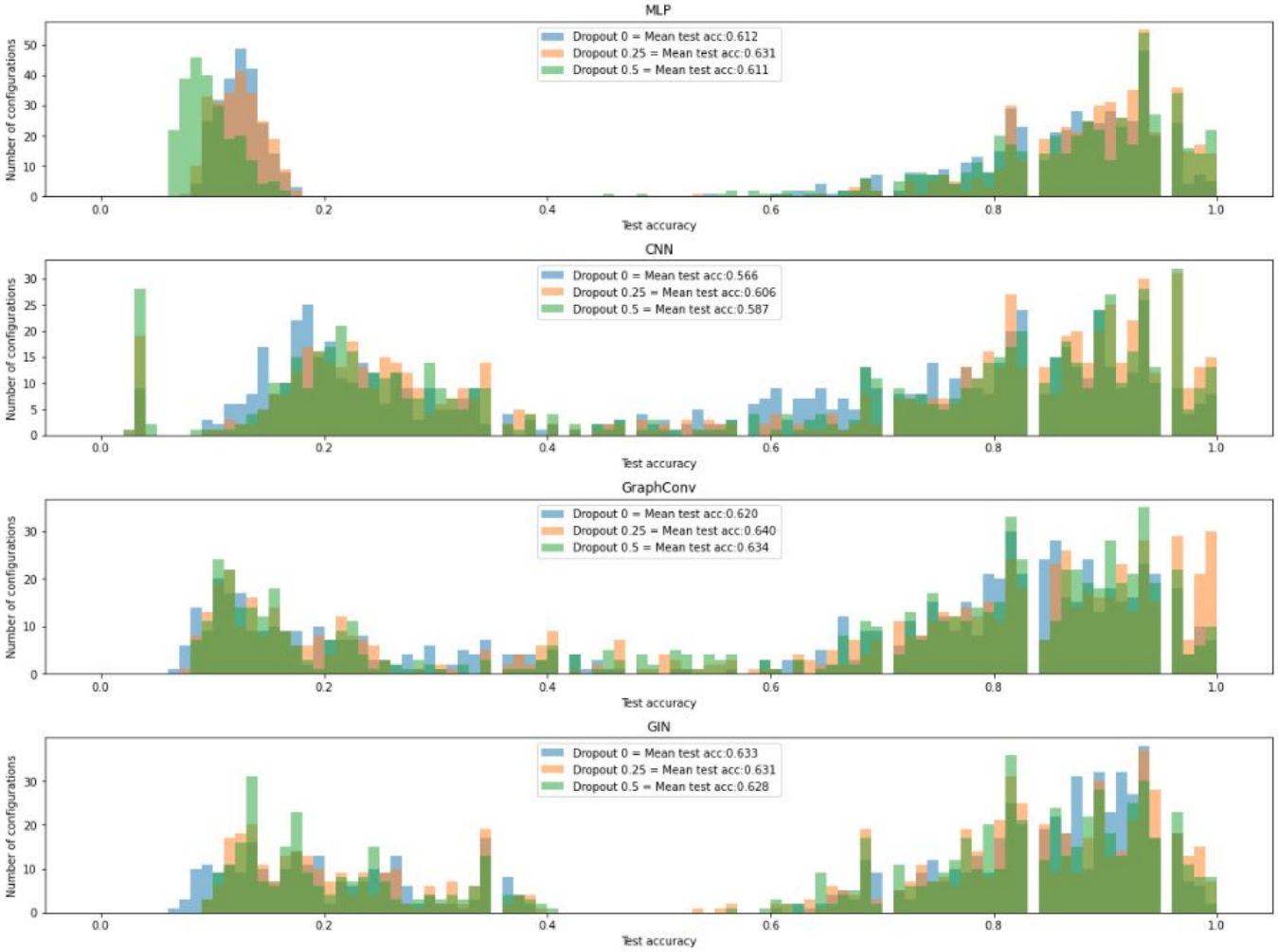
$$\mathbf{A}_{ij} = \mathbf{A}_{ij} - 1,$$

(2)

“ where (i, j) denotes the indices of global channel pairs: (FP1, FP2), (AF3, AF4), (F5, F6), (FC5, FC6), (C5, C6), (CP5, CP6), (P5, P6), (PO5, PO6) and (O1, O2) ”

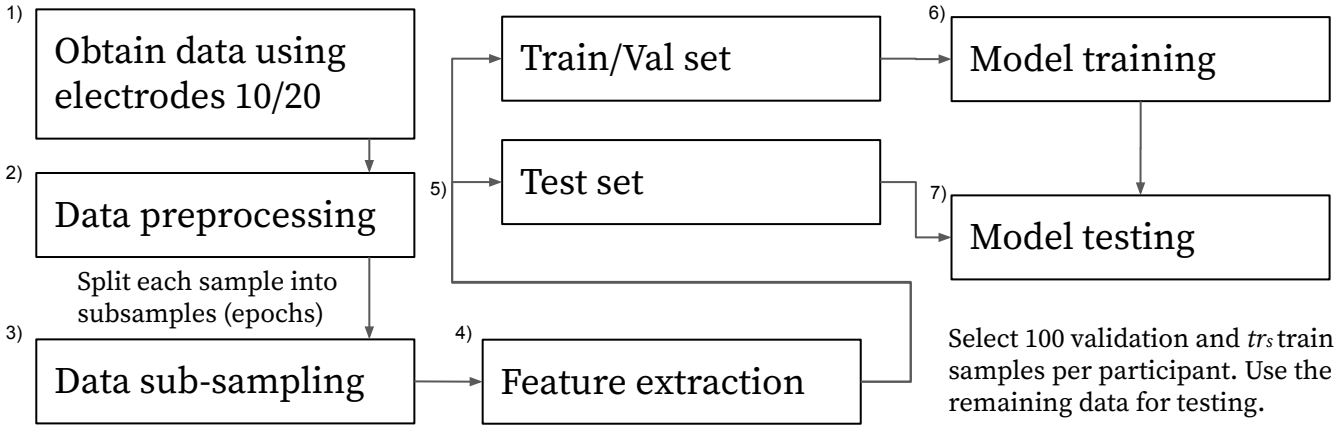
***** “δ > 0 denotes a calibration constant”
“δ = 5 such that around 20% of the entries in A are non-negligible”
“We empirically regard entries having values larger than 0.1 as non-negligible connections”

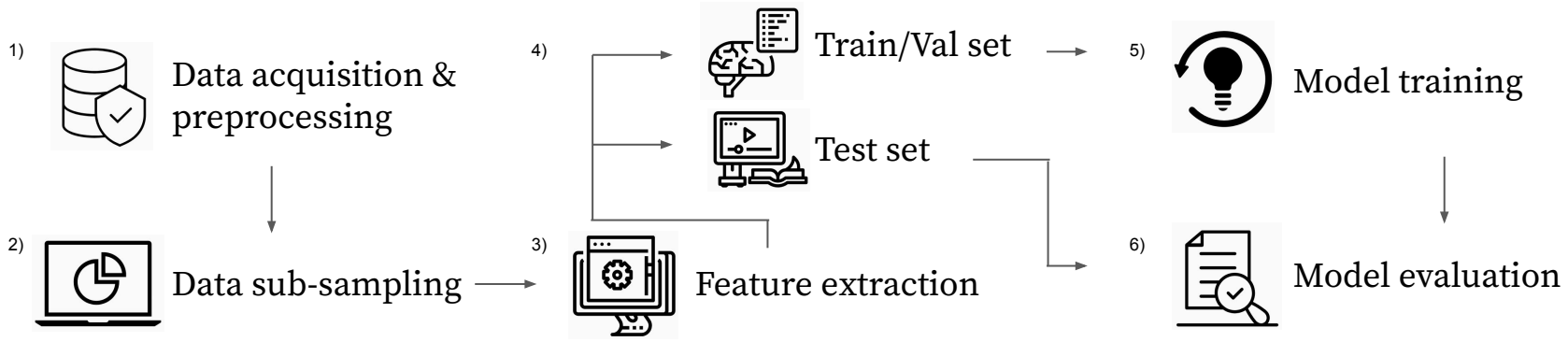
Zhong, P., Wang, D., & Miao, C. (2020). EEG-Based Emotion Recognition Using Regularized Graph Neural Networks. *IEEE Transactions on Affective Computing*, 1–1. https://doi.org/10.1109/TAFFC.2020.2994159


MLP
Dropout 0 = Mean test acc:0.612
Dropout 0.25 = Mean test acc:0.631
Dropout 0.5 = Mean test acc:0.611
Number of configurations
Test accuracy
CNN
Dropout 0 = Mean test acc:0.566
Dropout 0.25 = Mean test acc:0.606
Dropout 0.5 = Mean test acc:0.587
Number of configurations
Test accuracy
GraphConv
Dropout 0 = Mean test acc:0.620
Dropout 0.25 = Mean test acc:0.640
Dropout 0.5 = Mean test acc:0.634
Number of configurations
Test accuracy
GIN
Dropout 0 = Mean test acc:0.633
Dropout 0.25 = Mean test acc:0.631
Dropout 0.5 = Mean test acc:0.628
Number of configurations
Test accuracy

Experiment settings:

- Feature extraction method (wav,psd,raw)
- Epoch size (w)
- Number of train samples per participant (tr_s)
- Model architecture (LR, MLP, CNN, GraphConv)




1)
Data acquisition & preprocessing
2)
Data sub-sampling
3)
Feature extraction
4)
Train/Val set
Test set
5)
Model training
6)
Model evaluation